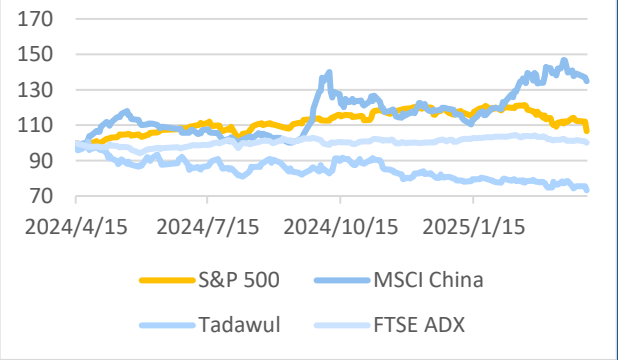


中东能源战略

油价突破每桶 105 美元，霍尔木兹海峡运输中断持续发酵

观点聚焦 Investment Focus

Name	Rating	Target		P/E	
		Price	2026E	2027E	
ADNOC Gas	Outperform	3.9	15.7	14.9	
ADNOC Drilling	Outperform	6.2	16.1	14.9	
ADNOC Distribution	Outperform	4.7	17.4	16.4	
ADNOC L&S	Outperform	7.0	11.5	10.9	
Borouge	Outperform	3.0	19.4	15.8	
Fertiglobe	Outperform	3.0	25.1	21.2	
Array Tech.	Outperform	11.8	11.2	9.4	
Ballard Power	Outperform	6.50	n.a.	n.a.	
Bloom Energy	Outperform	184	48.3	25.3	
Chart Industries	Outperform	227.0	12.1	9.7	
Clearway Energy	Outperform	34.0	38.8	27.4	
Industries Qatar	Outperform	15.5	15.6	14.1	
INPEX	Outperform	2,780	7.7	7.1	
Ma'aden	Outperform	58.0	31.1	29.1	
Nexttracker	Outperform	69.00	15.3	13.8	
Presight	Outperform	3.2	23.9	20.1	
solutions by stc	Outperform	359.0	18.9	17.2	
SABIC	Outperform	99.0	39.9	23.5	
Saudi Aramco	Outperform	32.8	16.4	15.8	
Vestas Wind Sys.	Outperform	131.00	18.6	13.3	
Enphase Energy	Neutral	85.00	17.4	13.9	
Plug Power	Neutral	5.00	n.a.	n.a.	
RENOVA	Neutral	1,450	20.6	19.3	
SolarEdge Tech.	Neutral	100.00	n.a.	n.a.	
JGC Holdings	Neutral	1,550	13.7	10.2	



Source: Factset, HTI

相关报告

中东能源战略：油价会再次飙升至每桶 100 美元吗？地缘冲突、霍尔木兹海峡与全球油价前景展望
中东能源战略：石油市场风险溢价重估
中东能源战略：中东地缘局势升级：对石油市场的影响
中东能源战略：油价飙升至每桶 110 美元以上：是什么将布伦特原油推至 120 美元-140 美元/桶？重新定价美国-伊朗紧张局势下的能源价值链

(This note is the Chinese translation of a note that was originally written in English and published on 16 Mar 2026)

中东冲突引发了一场规模空前的供应冲击。由于替代路线极为有限，加之陆上储油设施迅速满负荷，海湾产油国被迫关停产能（原油及其他液态燃料至少每日减产 1000 万桶）。若油轮运输无法迅速恢复，随着全系统运营瓶颈加剧，供应损失可能进一步扩大。

国际能源署预计 3 月全球石油供应将每日减少 800 万桶。尽管欧佩克+以外产油国增产以及哈萨克斯坦与俄罗斯此前中断的供应恢复正常，可部分抵消缺口，但仍不足以阻止净供应量的急剧下滑。除了短期冲击外，中期供需平衡仍显示 2026 年供应增长将超过需求增长，其中净增量基本全部来自非欧佩克+国家。

成品油供应紧张与原油停产共同加剧当前扰动。海湾地区成品油（2025 年日均 330 万桶）与液化石油气（日均 150 万桶）出口实际上已陷于停滞。受袭击事件及出口通道受阻影响，该地区已有逾每日 300 万桶的炼能关停，随着储罐接近容量极限，下行风险持续加剧。需求端亦现疲软——航班取消与石化原料短缺将使 3-4 月需求较预期每日减少 100 万桶，而价格波动率仍处高位。

鉴于全球观测库存达 82 亿桶，加之国际能源署授权紧急释放 4 亿桶战略储备，若霍尔木兹航运无法快速恢复，供应冲击影响将相对有限。

事件风险：哈尔克岛：哈尔克岛——作为伊朗海湾石油作业的核心枢纽，该岛被视为伊朗经济命脉。该岛原油出口终端名义吞吐能力达每日 700 万桶。若出口设施遭直接打击，伊朗每日 100 万-150 万桶的原油出口可能立即中断。

媒体报道显示，目前尚无石油基础设施受损的确切证据；美方官员称空袭针对军事目标，同时“规避石油设施。”但鉴于伊朗官员曾明确警告任何对能源基础设施的打击都将波及地区油气资产，主要风险依然高企——这预示着除伊朗出口外，更广泛的供应链与航运中断风险正在上升。

物流与保险成为核心制约：

短期油价定价逻辑已从“上游供应缺口”转向物流与保险约束。我们通过高频霍尔木兹海峡通行指标量化损失：离港油轮日均仅 6 艘，较正常情况的日均 70-80 艘锐减，意味着实际通过量仅为基础水平的 7.5%-8.6%（即常规流量损失约 90%）。在这种情况下，边际约束已从名义产能转向可交付供应——即实际能够装运、投保并完成交付的实物原油。

成功通航的油轮数量的减少从物理上收紧了即期供应，支撑近端溢价更具黏性、已实现/隐含波动率走高，以及曲线保持近端紧缩、远期定价相对疲软的态势。

我们的库存锚定与价格映射模型（SPR=120 天）

李丹怡 Catherine Li
catherine.dy.li@htisec.com

金雅皓 Zhihao Jin
zh.jin@htisec.com

在库存方面，我们以国际能源署的评估为锚定——即 3 月全球有效供应日均减少 800 万桶，并纳入由国际能源署主导的 32 国协同释放 4 亿桶战略石油储备（史上最大规模）这一因素。假设这 4 亿桶在 120 天内投放，则日均释放量约为 333 万桶。然而，鉴于运输能力仍严重受损，这部分释放转化为可交付的即期可交付实货的过程可能并不顺畅。

因此，在我们的核心假设下，预估短期净供应紧张程度依然较高，这与布伦特原油处于每桶 100 美元至 115 美元的高波动区间相吻合。

随着储备的投放、贸易流向的重新调整以及航运/保险可用性的部分恢复，净供应紧张程度和风险溢价应会边际减弱——这与未来 1 至 3 个月每桶 95 美元至 110 美元、3 个月后每桶 88 美元至 103 美元的油价水平相符。这一分析框架也解释了当前的市场结构（现货每桶 105 美元对 6 月期货每桶 100 美元）：近端价格由物流与承保能力等状态变量主导，而远端价格则受到储备释放路径及中期再平衡预期的制约。

情景 A（基准情形）：长期僵局

冲突持续但未显著升级。油轮通航量维持低位，保险市场保持紧张但仍有交易可能（高保费、限制性条款）。国际能源署估计的每日 800 万桶有效供应损失，其影响仍主要体现在曲线近端。

尽管宣布释放的 4 亿桶战略储备规模可观——按释放速度不同相当于每日 220 万-440 万桶——但紧张的运输与承保条件限制了这部分储备转化为即期可交付实货的能力。因此，短期净供应紧张程度难以显著缓解，仍将维持在高位区间。

价格影响（布伦特原油，美元/桶）曲线特征表现为近端风险溢价僵固，远端定价相对疲软：

1 个月内：每桶 100 美元-115 美元

1-3 个月：每桶 95 美元-110 美元

3 个月以上：每桶 88 美元-103 美元

情景 B（尾部风险）：冲突升级

重大袭击事件、军事对抗升级或石油航运直接受阻，将引发原油及成品油轮（包括液化天然气/液化石油气运输船）进一步撤离，同时承保能力再度收紧。

本就极低的通航活动要么进一步下降，要么在更长时间内“停滞在底部”（例如，每日过航船只数量持续停留在个位数），这意味着有效通航因子进一步恶化。在保险机制部分失效或完全失灵的情况下，战略储备向现货市场的传导不畅（即传递效率降低），推高净供应紧张程度，并引发风险溢价及已实现/隐含波动率的跳跃式上升。

价格影响（布伦特原油，美元/桶）

1 个月内：每桶 115 美元-135 美元（近端风险溢价涨幅最大）

1-3 个月：每桶 105 美元-125 美元（持续性取决于承保能力恢复速度，以及是否有可靠的护航或替代航线方案出现）

情景 C（下行风险）：局势降级/恢复正常

袭击事件显著减少；油轮通航量持续改善（例如，连续一周恢复），与石油相关的各类船舶稳定重返。伦敦海上保险公司与理赔协会的承保重新变得可行（即便保费仍处高位）。

关键在于并非单日反弹，而是通航因子的持久性回升，并伴随船舶类型的正常化。随着可交付性制约缓解，SPR 能更有效地转化为实际可供应的原油（尤其对于不依赖单一咽喉要道的物流通道而言）。净供应紧张程度与风险溢价将共同被重新定价至更低水平；曲线趋于平坦，近端跌幅大于远端。

价格影响（布伦特原油，美元/桶）

1 个月内：每桶 90 美元-102 美元

1-3 个月：每桶 85 美元-98 美元

3 个月以上：每桶 80 美元-92 美元

结构性转变：“付费通航”机制投入运行

一套正式的通航规则与费用/保证金机制开始实施——以基于规则的制度化成本，换取通航秩序的改善和承保能力的提升。随着船舶数量与船型组合恢复正常，通航因子上升；保险从一种依状态而定的尾部风险，转变为可定价的成本。这将压缩极端尾部风险，降低波动性，并收窄时间价差。然而，制度化的费用构成了价格下限，因此即使尾部风险消退，价格水平也可能不会大幅下跌。

价格影响（布伦特原油，美元/桶）

1 个月内：每桶 98 美元-112 美元

1- 3 个月：每桶 95 美元-108 美元

(与基准情形相比：中枢价格略强，但上行尾部风险得到更好控制)

情景 E（SPR）：E1 释放 vs. E2 恐慌

E1（SPR 主导的释放）

储备按节奏投放，切实改善了即期供应；现货升贴水收窄，曲线趋于平坦。

即使通航未能完全恢复正常，通过区域再平衡和替代效应，有效传递率也会显著提高，将价格拉向基准区间下沿，且对曲线远端的影响更为明显。

E2（“危机”恐慌）

如果通航持续“停滞在底部”（船舶数量长期停留在个位数），且事件发生频率依然很高，那么 SPR 行动将被解读为危机形势恶化的确认信号，从而引发买盘驱动的波动。

风险溢价短暂放大；近端波动率飙升；价格走势类似于情景 B 上行尾部风险的下半部分，但与真正、持续的实物供应中断相比，通常缺乏持续性。

1-4 周监测信号（动态情景）

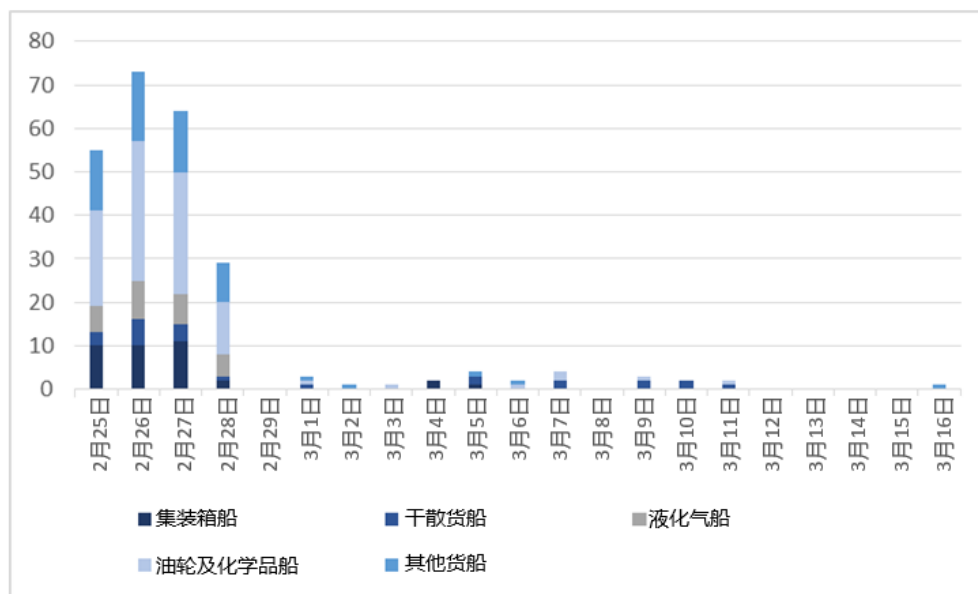
通航持续复苏（非单日数据）：每日船舶过航数量是否已连续一整周回升？原油/液化天然气/液化石油气相关船型是否在稳定返回？

保险机制转变：市场是否已从“无法承保”转向“可在高保费下交易”？这是区分离散的尾部风险跳跃与可定价成本之间的界限。

SPR 向即期库存的传导：现货升贴水是否收窄？曲线是否趋于平坦？这是否预示着在 120 天内释放 4 亿桶 SPR（约每日 330 万桶）正转化为可交付的供应？

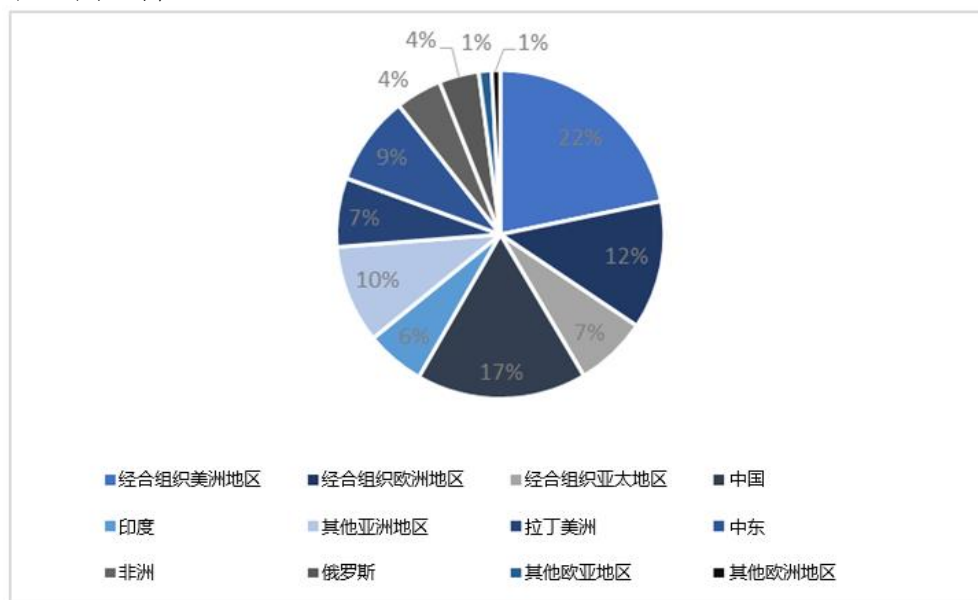
风险提示：新能源需求与价格走低；地缘政治风险

图 1 霍尔木兹海峡油品出港航运量



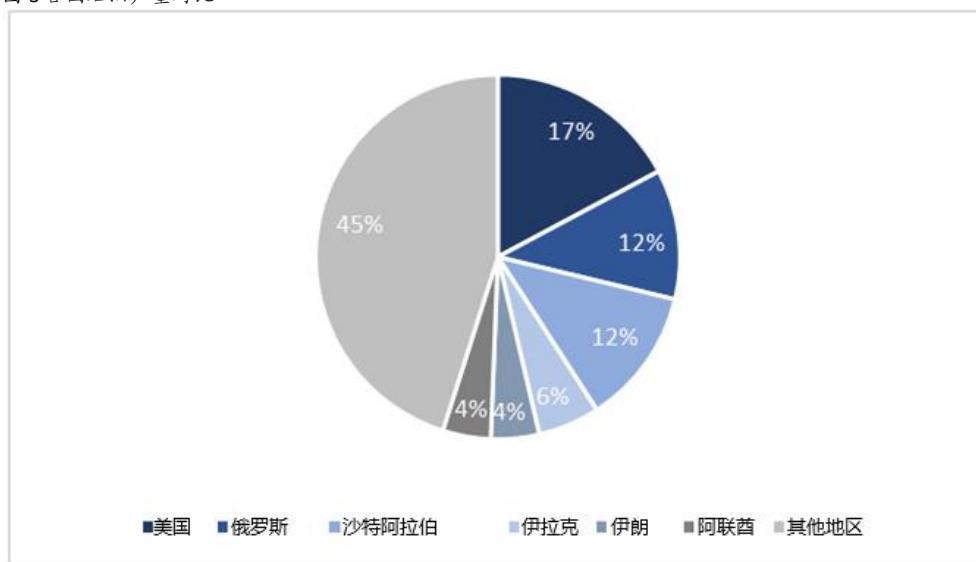
资料来源：HiFleet、海通国际

图 2 全球原油需求



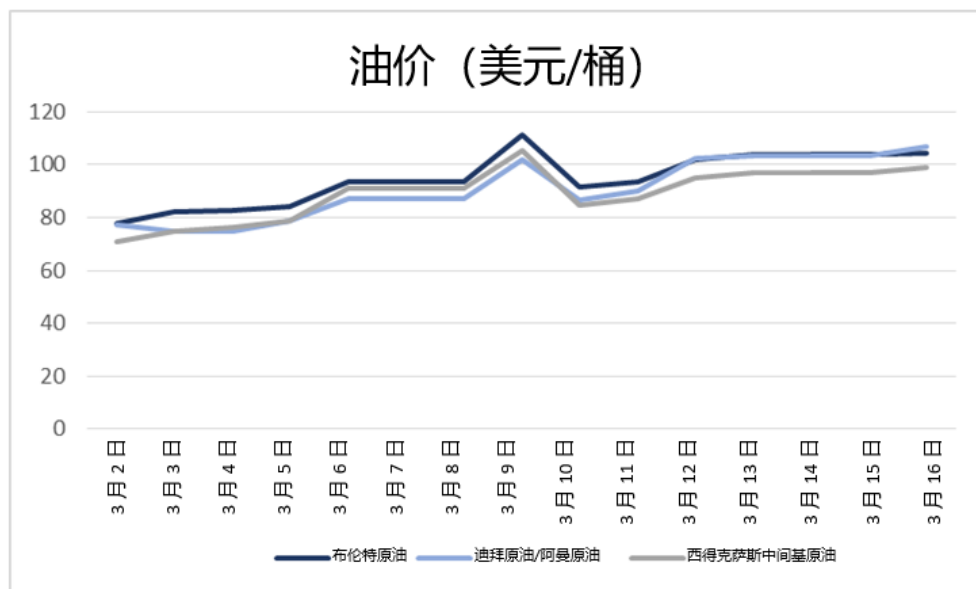
资料来源：欧佩克、海通国际

图 3 各国石油产量对比



资料来源：国际能源署、海通国际

图 4 油价



资料来源：彭博社、海通国际

周度价格	布伦特原油	西得克萨斯中间基原油	迪拜原油/阿曼原油平均	大庆	布伦特原油价格-西得克萨斯中间基原油	布伦特原油-迪拜原油	海通国际亚洲炼油毛利	汽油裂解价差	柴油裂解价差	航空煤油裂解价差	石脑油裂解价差	燃料油裂解价差	亨利枢纽	普氏日韩标杆指数	中国碳酸锂	氧化钨	尿素
2026年3月11日	87.2	83.3	86.6	91.4	3.9	0.6	35.7	31.6	55.9	54.2	7.4	(0.2)	3.0	15.9	22945.8	117641.1	560.0
2026年3月4日	81.4	74.7	78.3	80.5	6.7	3.1	41.1	14.2	55.0	139.7	5.8	(5.8)	2.9	15.1	22332.4	122900.9	491.0
2026年2月25日	70.9	65.4	70.5	70.5	5.4	0.4	10.1	6.6	20.9	21.7	(1.6)	(13.4)	3.0	10.6	21209.9	129008.8	452.5
2026年2月18日	70.4	65.2	68.3	68.7	5.2	2.0	8.4	4.6	15.5	15.4	(3.7)	(12.4)	3.0	10.4	19921.5	117641.1	452.5
周环比变化(%)																	
2026年3月11日	7.1%	11.6%	10.5%	13.6%	-42.9%	-80.1%	-13.2%	24.3%	4.9%	-36.1%	8.4%	15.1%	4.3%	5.4%	2.7%	-4.3%	14.1%
2026年3月4日	14.9%	14.1%	11.2%	14.2%	24.1%	665.0%	306.1%	23.5%	48.7%	139.0%	26.1%	31.5%	-1.8%	42.6%	5.3%	-4.7%	8.5%
2026年2月25日	0.7%	0.4%	3.1%	2.6%	5.2%	-80.4%	20.3%	3.3%	6.9%	7.9%	3.8%	-0.9%	-1.4%	1.9%	6.5%	9.7%	0.0%
月度价格至2026年3月	73.4	68.8	71.9	73.2	4.6	1.5	18.0	11.7	29.8	37.5	0.3	-10.3	3.5	12.2	15237.0	75036.4	466.6
2026年2月	86.6	81.7	83.4	86.5	4.8	3.2	35.7	22.2	52.5	74.3	4.1	-3.2	3.0	15.4	22815.5	121334.7	530.3
2026年1月	69.5	64.6	68.5	68.8	5.0	1.0	10.6	5.7	19.1	19.4	-1.3	-12.2	3.1	10.8	2082.4	11955.8	404
2025年12月	64.1	60.0	63.7	64.2	4.1	0.3	11.6	7.3	17.8	18.7	-2.0	-15.6	4.3	10.3	20813.2	91818.8	320.5
2025年11月	61.7	57.9	62.0	61.8	3.8	-0.3	12.6	14.0	19.2	22.7	0.1	-15.7	4.4	10.5	13944.8	82453.0	395.8
2025年10月	63.7	59.5	63.8	63.4	4.2	-0.2	13.6	14.9	28.0	29.2	0.7	-15.3	4.5	11.2	12134.8	77222.6	396.9
2025年9月	63.6	59.9	64.1	63.8	3.8	-0.4	14.6	12.7	22.4	22.4	0.0	-12.4	3.3	11.1	10334.9	75122.0	433.5
2025年8月	67.6	63.6	67.8	67.3	4.0	-0.2	15.6	12.1	20.8	19.6	-0.1	-12.9	3.0	11.3	10350.5	80935.5	433.5

2025 年 8 月	67.3	64.0	67.9	67.5	3.2	-0.6	16.6	12.8	23.7	17.2	-2.0	-12.3	2.9	11.6	11109.6	80223.4	476
2025 年 7 月	69.6	67.2	69.2	69.7	2.3	0.3	17.6	8.4	20.3	18.3	-4.1	-13.4	3.3	13.1	9432.7	67241.7	482.5
2025 年 6 月	69.8	67.3	68.5	69.1	2.5	1.3	18.6	10.2	19.8	17.8	-4.5	-13.7	3.3	13.2	9377.1	66728.7	384.3
2025 年 5 月	75.7	71.6	75.1	71.8	4.1	0.7	19.6	9.9	12.3	13.1	-8.6	-18.4	2.3	10.5	34906.7	67014.4	469.3
2025 年 4 月	83.4	79.5	83.8	81.0	3.9	0.1	20.6	11.3	13.3	14.2	-7.1	-21.6	2.2	12.3	27601.1	70541.3	417.8
2025 年 3 月	79.2	73.4	77.5	75.8	5.8	0.8	21.6	14.3	21.0	20.3	-1.9	-22.8	2.4	13.6	44443.0	84563.2	374.5
2025 年 2 月	83.5	76.9	81.2	79.1	6.7	1.8	22.6	11.3	20.3	20.9	-1.8	-28.3	2.4	16.9	63363.9	103697.4	398
2025 年 1 月	84.0	78.2	80.0	80.0	5.8	3.4	23.6	11.2	29.7	30.2	-6.9	-30.4	3.4	24.3	72114.4	106537.1	353.1
2024 年 12 月	81.3	76.5	76.8	79.3	4.8	4.3	24.6	3.5	29.5	30.5	-8.1	-29.5	5.8	32.3	79524.9	99095.1	319.9
2024 年 11 月	90.9	84.4	86.3	91.0	6.5	4.8	25.6	-0.9	30.3	28.5	-12.5	-35.7	6.4	28.4	81834.2	90668.0	319.9
2024 年 10 月	93.6	87.0	90.6	92.2	6.6	1.7	26.6	-3.2	35.7	29.5	-15.8	-41.0	6.1	33.0	75453.8	93612.7	319.9
月环比变化 (%)																	
2026 年 3 月	24.5%	26.6%	21.7%	25.7%	-2.5%	221.2%	237.5%	293.2%	175.1%	284.0%	-424.8%	-73.5%	-2.5%	42.7%	995.6%	914.9%	31%
2026 年 2 月	8.5%	7.7%	7.5%	7.2%	20.8%	193.4%	-8.6%	-23.0%	7.2%	3.6%	-36.5%	-21.7%	-26.8%	5.3%	-90.0%	-87.0%	26%
2026 年 1 月	3.9%	3.6%	2.8%	3.8%	7.9%	-202.8%	-7.9%	-47.7%	-7.1%	-17.8%	-2681.2%	-0.5%	-3.5%	-1.9%	49.3%	11.4%	-19%
2025 年 12 月	-3.1%	-2.7%	-2.8%	-2.5%	-9.1%	99.7%	-7.4%	-5.7%	-31.6%	-22.1%	-88.8%	2.1%	-1.2%	-6.8%	14.9%	6.8%	0%
2025 年 11 月	0.0%	-0.7%	-0.4%	-0.7%	10.9%	-59.6%	-6.9%	17.2%	25.2%	30.3%	-2241.0%	24.1%	36.9%	1.2%	17.4%	2.8%	-8%
2025 年 10 月	-5.8%	-5.9%	-5.5%	-5.2%	-5.1%	137.4%	-6.4%	5.3%	7.4%	14.0%	-52.2%	-4.0%	8.0%	-1.8%	-0.2%	-7.2%	0%
2025 年 9 月	0.5%	-0.6%	-0.2%	-0.2%	22.5%	-71.6%	-6.0%	-5.9%	-12.1%	14.4%	-96.6%	5.1%	4.6%	-2.7%	-6.8%	0.9%	-9%

2025 年 8 月	-3.3%	-4.8%	-2.0%	-3.2%	40.3%	-285.7%	-5.7%	53.4%	16.8%	-6.1%	-52.1%	-8.9%	-12.5%	-11.2%	17.8%	19.3%	-1%
2025 年 7 月	-0.4%	-0.1%	1.1%	0.8%	-6.7%	-75.0%	-5.4%	-18.4%	2.6%	2.8%	-9.5%	-1.8%	-0.5%	-0.6%	0.6%	0.8%	26%
2025 年 6 月	-7.8%	-6.0%	-8.8%	-3.8%	-39.2%	96.4%	-5.1%	3.6%	60.6%	36.2%	-47.3%	-25.7%	44.3%	26.1%	-73.1%	-0.4%	-18%
2025 年 5 月	-9.2%	-9.9%	-10.4%	-11.3%	5.3%	355.1%	-4.9%	-12.6%	-7.5%	-7.9%	22.1%	-14.7%	5.1%	-15.3%	26.5%	-5.0%	12%
2025 年 4 月	5.3%	8.4%	8.2%	6.8%	-33.7%	-82.8%	-4.6%	-20.9%	-36.6%	-30.1%	268.0%	-5.3%	-9.1%	-9.2%	-37.9%	-16.6%	12%
2025 年 3 月	-5.2%	-4.5%	-4.6%	-4.2%	-12.6%	-51.9%	-4.4%	26.7%	3.4%	-3.2%	5.7%	-19.4%	-1.2%	-19.4%	-29.9%	-18.5%	-6%
2025 年 2 月	-0.5%	-1.7%	1.5%	-1.2%	14.5%	-47.5%	-4.2%	0.5%	-31.6%	-30.6%	-73.6%	-6.8%	-28.8%	-30.7%	-12.1%	-2.7%	13%
2025 年 1 月	3.3%	2.2%	4.2%	0.9%	20.9%	-21.6%	-4.1%	222.6%	0.6%	-1.2%	-14.9%	2.8%	-40.7%	-24.7%	-9.3%	7.5%	10%
2024 年 12 月	-10.5%	-9.3%	-11.0%	-12.8%	-25.4%	-10.2%	-3.9%	-506.5%	-2.8%	6.9%	-35.8%	-17.2%	-10.3%	14.0%	-2.8%	9.3%	0%
2024 年 11 月	-2.9%	-3.0%	-4.8%	-1.3%	-1.5%	188.3%	-3.8%	-73.0%	-15.1%	-3.3%	-20.7%	-12.9%	5.7%	-14.0%	8.5%	-3.1%	0%

资料来源：彭博社、海通国际测算。价格截至 2026 年 3 月 11 日。所有石油、石油产品裂解价差单位为美元/桶，所有天然气价格单位为美元/百万英热单位。HH=亨利枢纽、NBP=NBP、JKM=普氏日韩标杆指数（运往日本、韩国、中国及台湾地区的液化天然气现货基准价格）。中国碳酸锂、氧化钨、尿素以及聚乙烯-石脑油的价格单位为美元/公吨。

附录 APPENDIX

重要信息披露

本研究报告由海通国际分销，海通国际是由海通国际研究有限公司 (HTIRL)，Haitong Securities India Private Limited (HSIPL)，Haitong International Japan K.K. (HTIJKK) 和海通国际证券有限公司 (HTISCL) 的证券研究团队所组成的全球品牌，海通国际证券集团 (HTISG) 各成员分别在其许可的司法管辖区内从事证券活动。

IMPORTANT DISCLOSURES

This research report is distributed by Haitong International, a global brand name for the equity research teams of Haitong International Research Limited ("HTIRL"), Haitong Securities India Private Limited ("HSIPL"), Haitong International Japan K.K. ("HTIJKK"), Haitong International Securities Company Limited ("HTISCL"), and any other members within the Haitong International Securities Group of Companies ("HTISG"), each authorized to engage in securities activities in its respective jurisdiction.

HTIRL 分析师认证 Analyst Certification:

我，李丹怡，在此保证 (i) 本研究报告中的意见准确反映了我们对本研究中提及的任何或所有目标公司或上市公司的个人观点，并且 (ii) 我的报酬中没有任何部分与本研究报告中表达的具体建议或观点直接或间接相关；及就此报告中所讨论目标公司的证券，我们（包括我们的家属）在其中均不持有任何财务利益。我和我的家属（我已经告知他们）将不会在本研究报告发布后的 30 个自然日内交易此研究报告所讨论目标公司的证券。I, Catherine Li, certify that (i) the views expressed in this research report accurately reflect my personal views about any or all of the subject companies or issuers referred to in this research and (ii) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report; and that I (including members of my household) have no financial interest in the security or securities of the subject companies discussed. I and my household, whom I have already notified of this, will not deal in or trade any securities in respect of the issuer that I review within 30 calendar days after the research report is published.

我，金稚皓，在此保证 (i) 本研究报告中的意见准确反映了我们对本研究中提及的任何或所有目标公司或上市公司的个人观点，并且 (ii) 我的报酬中没有任何部分与本研究报告中表达的具体建议或观点直接或间接相关；及就此报告中所讨论目标公司的证券，我们（包括我们的家属）在其中均不持有任何财务利益。我和我的家属（我已经告知他们）将不会在本研究报告发布后的 30 个自然日内交易此研究报告所讨论目标公司的证券。I, Zhihao Jin, certify that (i) the views expressed in this research report accurately reflect my personal views about any or all of the subject companies or issuers referred to in this research and (ii) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report; and that I (including members of my household) have no financial interest in the security or securities of the subject companies discussed. I and my household, whom I have already notified of this, will not deal in or trade any securities in respect of the issuer that I review within 30 calendar days after the research report is published.

利益冲突披露 Conflict of Interest Disclosures

海通国际及其某些关联公司可从事投资银行业务和/或对本研究中的特定股票或公司进行做市或持有自营头寸。就本研究报告而言，以下是有关该等关系的披露事项（以下披露不能保证及时无遗漏，如需了解及时全面信息，请发邮件至 ERD-Disclosure@htisec.com）

HTI and some of its affiliates may engage in investment banking and / or serve as a market maker or hold proprietary trading positions of certain stocks or companies in this research report. As far as this research report is concerned, the following are the disclosure matters related to such relationship (As the following disclosure does not ensure timeliness and completeness, please send an email to ERD-Disclosure@htisec.com if timely and comprehensive information is needed).

国泰海通证券股份有限公司和/或其子公司（统称“国泰海通”）在过去 12 个月内参与了 600028.CH, 600346.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK 的投资银行项目。投资银行项目包括：1、国泰海通担任上市前辅导机构、保荐人或主承销商的首次公开发行项目；2、国泰海通作为保荐人、主承销商或财务顾问的股权或债务再融资项目；3、国泰海通作为主经纪商的新三板上市、目标配售和并购项目。

Guotai Haitong Securities Co., Ltd. and/or its subsidiaries (collectively, the "Guotai Haitong") have a role in investment banking projects of 600028.CH, 600346.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK within the past 12 months. The investment banking projects include 1. IPO projects in which Guotai Haitong acted as pre-listing tutor, sponsor, or lead-underwriter; 2. equity or debt refinancing projects of 600028.CH, 600346.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK for which Guotai Haitong acted as sponsor, lead-underwriter or financial advisor; 3. listing by introduction in the new three board, target placement, M&A projects in which Guotai Haitong acted as lead-brokerage firm.

600028.CH, 600346.CH, 600688.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH 及 2727.HK 目前或过去 12 个月内是国泰海通的投资银行业务客户。

600028.CH, 600346.CH, 600688.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK are/were an investment bank clients of Guotai Haitong currently or within the past 12 months.

002335.CH, 600023.CH, 002028.CH, 605090.CH, 002074.CH, 2727.HK 及 600256.CH 目前或过去 12 个月内是国泰海通的客户。国泰海通向客户提供非投资银行业务的证券相关业务服务。

002335.CH, 600023.CH, 002028.CH, 605090.CH, 002074.CH, 2727.HK and 600256.CH are/were a client of Guotai Haitong currently or within the past 12 months. The client has been provided for non-investment-banking securities-related services.

国泰海通在过去 12 个月中获得对 600028.CH, 600346.CH, 600688.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH 及 2727.HK 提供投资银行服务的报酬。

Guotai Haitong received in the past 12 months compensation for investment banking services provided to 600028.CH, 600346.CH, 600688.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK.

国泰海通预计将（或者有意向）在未来三个月内从 600028.CH, 600346.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH 及 2727.HK 获得投资银行服务报酬。

Guotai Haitong expects to receive, or intends to seek, compensation for investment banking services in the next three months from 600028.CH, 600346.CH, 0857.HK, 601808.CH, 600803.CH, 300014.CH, 002074.CH and 2727.HK.

国泰海通在过去的 12 个月中从 600256.CH 获得除投资银行服务以外之产品或服务的报酬。

Guotai Haitong has received compensation in the past 12 months for products or services other than investment banking from 600256.CH.

国泰海通担任 688248.CH, 0857.HK 及 0883.HK 有关证券的做市商或流通量提供者。

Guotai Haitong acts as a market maker or liquidity provider in the securities of 688248.CH, 0857.HK and 0883.HK.

评级定义 (从 2020 年 7 月 1 日开始执行):

海通国际 (以下简称“HTI”) 采用相对评级系统来为投资者推荐我们覆盖的公司: 优于大市、中性或弱于大市。投资者应仔细阅读 HTI 的评级定义。并且 HTI 发布分析师观点的完整信息, 投资者应仔细阅读全文而非仅看评级。在任何情况下, 分析师的评级和研究都不能作为投资建议。投资者的买卖股票的决策应基于各自情况 (比如投资者的现有持仓) 以及其他因素。

分析师股票评级

优于大市, 未来 12-18 个月内预期相对基准指数涨幅在 10%以上, 基准定义如下

中性, 未来 12-18 个月内预期相对基准指数变化不大, 基准定义如下。根据 FINRA/NYSE 的评级分布规则, 我们会将中性评级划入持有这一类别。

弱于大市, 未来 12-18 个月内预期相对基准指数跌幅在 10%以上, 基准定义如下

各地股票基准指数: 日本 – TOPIX, 韩国 – KOSPI, 台湾 – TAIEX, 印度 – Nifty100, 美国 – SP500; 其他所有中国概念股 – MSCI China.

Ratings Definitions (from 1 Jul 2020):

Haitong International uses a relative rating system using Outperform, Neutral, or Underperform for recommending the stocks we cover to investors. Investors should carefully read the definitions of all ratings used in Haitong International Research. In addition, since Haitong International Research contains more complete information concerning the analyst's views, investors should carefully read Haitong International Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Analyst Stock Ratings

Outperform: The stock's total return over the next 12-18 months is expected to exceed the return of its relevant broad market benchmark, as indicated below.

Neutral: The stock's total return over the next 12-18 months is expected to be in line with the return of its relevant broad market benchmark, as indicated below. For purposes only of FINRA/NYSE ratings distribution rules, our Neutral rating falls into a hold rating category.

Underperform: The stock's total return over the next 12-18 months is expected to be below the return of its relevant broad market benchmark, as indicated below.

Benchmarks for each stock's listed region are as follows: Japan – TOPIX, Korea – KOSPI, Taiwan – TAIEX, India – Nifty100, US – SP500; for all other China-concept stocks – MSCI China.

	截至 2025 年 12 月 31 日海通国际股票研究评级分布			截至 2025 年 9 月 30 日海通国际股票研究评级分布		
	优于大市	中性 (持有)	弱于大市	优于大市	中性 (持有)	弱于大市
海通国际股票研究覆盖率	93.9%	6.0%	0.1%	92.3%	7.5%	0.2%
投资银行客户*	3.0%	4.0%	0.0%	3.3%	3.9%	0.0%

*在每个评级类别里投资银行客户所占的百分比。

上述分布中的买入, 中性和卖出分别对应我们当前优于大市, 中性和落后大市评级。

只有根据 FINRA/NYSE 的评级分布规则, 我们才将中性评级划入持有这一类别。请注意在上表中不包含非评级的股票。

此前的评级系统定义 (直至 2020 年 6 月 30 日):

买入, 未来 12-18 个月内预期相对基准指数涨幅在 10%以上, 基准定义如下

中性, 未来 12-18 个月内预期相对基准指数变化不大, 基准定义如下。根据 FINRA/NYSE 的评级分布规则, 我们会将中性评级划入持有这一类别。

卖出, 未来 12-18 个月内预期相对基准指数跌幅在 10%以上, 基准定义如下

各地股票基准指数: 日本 – TOPIX, 韩国 – KOSPI, 台湾 – TAIEX, 印度 – Nifty100; 其他所有中国概念股 – MSCI China.

	Haitong International Equity Research Ratings Distribution, as of December 31, 2025			Haitong International Equity Research Ratings Distribution, as of September 30, 2025		
	Outperform	Neutral (hold)	Underperform	Outperform	Neutral (hold)	Underperform
HTI Equity Research Coverage	93.9%	6.0%	0.1%	92.3%	7.5%	0.2%
IB clients*	3.0%	4.0%	0.0%	3.3%	3.9%	0.0%

*Percentage of investment banking clients in each rating category.

BUY, Neutral, and SELL in the above distribution correspond to our current ratings of Outperform, Neutral, and Underperform.

For purposes only of FINRA/NYSE ratings distribution rules, our Neutral rating falls into a hold rating category. Please note that stocks with an NR designation are not included in the table above.

Previous rating system definitions (until 30 Jun 2020):

BUY: The stock's total return over the next 12-18 months is expected to exceed the return of its relevant broad market benchmark, as indicated below.

NEUTRAL: The stock's total return over the next 12-18 months is expected to be in line with the return of its relevant broad market benchmark, as indicated below. For purposes only of FINRA/NYSE ratings distribution rules, our Neutral rating falls into a hold rating category.

SELL: The stock's total return over the next 12-18 months is expected to be below the return of its relevant broad market benchmark, as indicated below.

Benchmarks for each stock's listed region are as follows: Japan – TOPIX, Korea – KOSPI, Taiwan – TAIEX, India – Nifty100; for all other China-concept stocks – MSCI China.

海通国际非评级研究: 海通国际发布计量、筛选或短篇报告，并在报告中根据估值和其他指标对股票进行排名，或者基于可能的估值倍数提出建议价格。这种排名或建议价格并非为

Haitong International Non-Rated Research: Haitong International publishes quantitative, screening or short reports which may rank stocks according to valuation and other metrics or may suggest prices based on possible valuation multiples. Such rankings or suggested prices do not purport to be stock ratings or target prices or fundamental values and are for information only.

海通国际 A 股覆盖: 海通国际可能会就沪港通及深港通的中国 A 股进行覆盖及评级。国泰海通证券（601211.CH），海通国际于上海的母公司，也会于中国发布中国 A 股的研究报告。但是，海通国际使用与国泰海通证券不同的评级系统，所以海通国际与国泰海通证券的中国 A 股评级可能有所不同。

Haitong International Coverage of A-Shares: Haitong International may cover and rate A-Shares that are subject to the Hong Kong Stock Connect scheme with Shanghai and Shenzhen. GUOTAI HAITONG SECURITIES (601211 CH), the ultimate parent company of HTISG based in Shanghai, covers and publishes research on these same A-Shares for distribution in mainland China. However, the rating system employed by GTHS differs from that used by HTI and as a result there may be a difference in the HTI and GTHS ratings for the same A-share stocks.

海通国际优质 100 A 股 (Q100) 指数: 海通国际 Q100 指数是一个包括 100 支由国泰海通证券覆盖的优质中国 A 股的计量产品。这些股票是通过基于质量的筛选过程，并结合对国泰海通证券 A 股团队自下而上的研究。海通国际每季对 Q100 指数成分作出复审。

Haitong International Quality 100 A-share (Q100) Index: HTI's Q100 Index is a quant product that consists of 100 of the highest-quality A-shares under coverage at GTHS in Shanghai. These stocks are carefully selected through a quality-based screening process in combination with a review of the GTHS A-share team's bottom-up research. The Q100 constituent companies are reviewed quarterly.

盟浪义利 (FIN-ESG) 数据通免责声明条款: 在使用盟浪义利 (FIN-ESG) 数据之前，请务必仔细阅读本条款并同意本声明：

第一条 义利 (FIN-ESG) 数据系由盟浪可持续数字科技有限责任公司（以下简称“本公司”）基于合法取得的公开信息评估而成，本公司对信息的准确性及完整性不作任何保证。对公司的评估结果仅供参考，并不构成对任何个人或机构投资建议，也不能作为任何个人或机构购买、出售或持有相关金融产品的依据。本公司不对任何个人或机构投资者因使用本数据表述的评估结果造成的任何直接或间接损失负责。

第二条 盟浪并不因此收到此评估数据而将收件人视为客户，收件人使用此数据时应根据自身实际情况作出自我独立判断。本数据所载内容反映的是盟浪在最初发布本数据日期当日的判断，盟浪有权在不发出通知的情况下更新、修订与发出其他与本数据所载内容不一致或有不同结论的数据。除非另行说明，本数据（如财务业绩数据等）仅代表过往表现，过往的业绩表现不作为日后回报的预测。

第三条 本数据版权归本公司所有，本公司依法保留各项权利。未经本公司事先书面许可授权，任何个人或机构不得将本数据中的评估结果用于任何营利性目的，不得对本数据进行修改、复制、编译、汇编、再次编辑、改编、删减、缩写、节选、发行、出租、展览、表演、放映、广播、信息网络传播、摄制、增加图标及说明等，否则因此给盟浪或其他第三方造成损失的，由用户承担相应的赔偿责任，盟浪不承担责任。

第四条 如本免责声明未约定，而盟浪网站平台载明的其他协议内容（如《盟浪网站用户注册协议》《盟浪网用户服务（含认证）协议》《盟浪网隐私政策》等）有约定的，则按其他协议的约定执行；若本免责声明与其他协议约定存在冲突或不一致的，则以本免责声明约定为准。

SusallWave FIN-ESG Data Service Disclaimer: Please read these terms and conditions below carefully and confirm your agreement and acceptance with these terms before using SusallWave FIN-ESG Data Service.

1. FIN-ESG Data is produced by SusallWave Digital Technology Co., Ltd. (In short, SusallWave)'s assessment based on legal publicly accessible information. SusallWave shall not be responsible for any accuracy and completeness of the information. The assessment result is for reference only. It is not for any investment advice for any individual or institution and not for basis of purchasing, selling or holding any relative financial products. We will not be liable for any direct or indirect loss of any individual or institution as a result of using SusallWave FIN-ESG Data.

2. SusallWave do not consider recipients as customers for receiving these data. When using the data, recipients shall make your own independent judgment according to your practical individual status. The contents of the data reflect the judgment of us only on the release day. We have right to update and amend the data and release other data that contains inconsistent contents or different conclusions without notification. Unless expressly stated, the data (e.g., financial performance data) represents past performance only and the past performance cannot be viewed as the prediction of future return.

3. The copyright of this data belongs to SusallWave, and we reserve all rights in accordance with the law. Without the prior written permission of our company, none of individual or institution can use these data for any profitable purpose. Besides, none of individual or institution can take actions such as amendment, replication, translation, compilation, re-editing, adaption, deletion, abbreviation, excerpts, issuance, rent, exhibition, performance, projection, broadcast, information network transmission, shooting, adding icons and instructions. If any loss of SusallWave or any third-party is caused by those actions, users shall bear the corresponding compensation liability. SusallWave shall not be responsible for any loss.

4. If any term is not contained in this disclaimer but written in other agreements on our website (e.g. *User Registration Protocol of SusallWave Website*, *User Service (including authentication) Agreement of SusallWave Website*, *Privacy Policy of Susallwave Website*), it should be executed according to other agreements. If there is any difference between this disclaimer and other agreements, this disclaimer shall be applied.

重要免责声明:

非印度证券的研究报告: 本报告由海通国际证券集团有限公司（“HTISGL”）的全资附属公司海通国际研究有限公司（“HTIRL”）发行，该公司是根据香港证券及期货条例（第 571 章）持有第 4 类受规管活动（就证券提供意见）的持牌法团。该研究报告在 HTISGL 的全资附属公司 Haitong International (Japan) K.K.（“HTIJK”）的协助下发行，HTIJK 是由日本关东财务局监管为投资顾问。

印度证券的研究报告：本报告由从事证券交易、投资银行及证券分析及受 Securities and Exchange Board of India (“SEBI”) 监管的 Haitong Securities India Private Limited (“HTSIPL”) 所发行，包括制作及发布涵盖 BSE Limited (“BSE”) 和 National Stock Exchange of India Limited (“NSE”) 上市公司（统称为「印度交易所」）的研究报告。HTSIPL 于 2016 年 12 月 22 日被收购并成为海通国际证券集团有限公司 (“HTISG”) 的一部分。

所有研究报告均以海通国际为名作为全球品牌，经许可由海通国际证券股份有限公司及/或海通国际证券集团的其他成员在其司法管辖区发布。

本文件所载信息和观点已被编译或源自可靠来源，但 HTIRL、HTISCL 或任何其他属于海通国际证券集团有限公司 (“HTISG”) 的成员对其准确性、完整性和正确性不做任何明示或暗示的声明或保证。本文件中所有观点均截至本报告日期，如有更改，恕不另行通知。本文件仅供参考使用。文件中提及的任何公司或其股票的说明并非意图展示完整的内容，本文件并非/不应被解释为对证券买卖的明示或暗示地出价或征价。在某些司法管辖区，本文件中提及的证券可能无法进行买卖。如果投资产品以投资者本国货币以外的币种进行计价，则汇率变化可能会对投资产生不利影响。过去的表现并不一定代表将来的结果。某些特定交易，包括设计金融衍生工具的，有产生重大风险的可能性，因此并不适合所有的投资者。您还应认识到本文件中的建议并非为您量身定制。分析师并未考虑到您自身的财务情况，如您的财务状况和风险偏好。因此您必须自行分析并在适用的情况下咨询自己的法律、税收、会计、金融和其他方面的专业顾问，以期在投资之前评估该项建议是否适合于您。若由于使用本文件所载的材料而产生任何直接或间接的损失，HTISG 及其董事、雇员或代理人对此均不承担任何责任。

除对本文内容承担责任的分析师除外，HTISG 及我们的关联公司、高级管理人员、董事和雇员，均可不时作为主事人就本文件所述的任何证券或衍生品持有长仓或短仓以及进行买卖。HTISG 的销售员、交易员和其他专业人士均可向 HTISG 的相关客户和公司提供与本文件所述意见相反的口头或书面市场评论意见或交易策略。HTISG 可做出与本文件所述建议或意见不一致的投资决策。但 HTIRL 没有义务来确保本文件的收件人了解到该等交易决定、思路或建议。

请访问海通国际网站 www.equities.htisec.com，查阅更多有关海通国际为预防和避免利益冲突设立的组织和行政安排的内容信息。

非美国分析师披露信息：本项研究首页上列明的海通国际分析师并未在 FINRA 进行注册或者取得相应的资格，并且不受美国 FINRA 有关与本项研究目标公司进行沟通、公开露面和自售证券交易的第 2241 条规则之限制。

IMPORTANT DISCLAIMER

For research reports on non-Indian securities: The research report is issued by Haitong International Research Limited (“HTIRL”), a wholly owned subsidiary of Haitong International Securities Group Limited (“HTISGL”) and a licensed corporation to carry on Type 4 regulated activity (advising on securities) for the purpose of the Securities and Futures Ordinance (Cap. 571) of Hong Kong, with the assistance of Haitong International (Japan) K.K. (“HTIJKK”), a wholly owned subsidiary of HTISGL and which is regulated as an Investment Adviser by the Kanto Finance Bureau of Japan.

For research reports on Indian securities: The research report is issued by Haitong Securities India Private Limited (“HTSIPL”), an Indian company and a Securities and Exchange Board of India (“SEBI”) registered Stock Broker, Merchant Banker and Research Analyst that, inter alia, produces and distributes research reports covering listed entities on the BSE Limited (“BSE”) and the National Stock Exchange of India Limited (“NSE”) (collectively referred to as “Indian Exchanges”). HTSIPL was acquired and became part of the Haitong International Securities Group of Companies (“HTISG”) on 22 December 2016.

All the research reports are globally branded under the name Haitong International and approved for distribution by Haitong International Securities Company Limited (“HTISCL”) and/or any other members within HTISG in their respective jurisdictions.

The information and opinions contained in this research report have been compiled or arrived at from sources believed to be reliable and in good faith but no representation or warranty, express or implied, is made by HTIRL, HTISCL, HTSIPL, HTIJKK or any other members within HTISG from which this research report may be received, as to their accuracy, completeness or correctness. All opinions expressed herein are as of the date of this research report and are subject to change without notice. This research report is for information purpose only. Descriptions of any companies or their securities mentioned herein are not intended to be complete and this research report is not, and should not be construed expressly or impliedly as, an offer to buy or sell securities. The securities referred to in this research report may not be eligible for purchase or sale in some jurisdictions. If an investment product is denominated in a currency other than an investor's home currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Certain transactions, including those involving derivatives, give rise to substantial risk and are not suitable for all investors. You should also bear in mind that recommendations in this research report are not tailor-made for you. The analyst has not taken into account your unique financial circumstances, such as your financial situation and risk appetite. You must, therefore, analyze and should, where applicable, consult your own legal, tax, accounting, financial and other professional advisers to evaluate whether the recommendations suits you before investment. Neither HTISG nor any of its directors, employees or agents accepts any liability whatsoever for any direct or consequential loss arising from any use of the materials contained in this research report.

HTISG and our affiliates, officers, directors, and employees, excluding the analysts responsible for the content of this document, will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research report. Sales, traders, and other professionals of HTISG may provide oral or written market commentary or trading strategies to the relevant clients and the companies within HTISG that reflect opinions that are contrary to the opinions expressed in this research report. HTISG may make investment decisions that are inconsistent with the recommendations or views expressed in this research report. HTI is under no obligation to ensure that such other trading decisions, ideas or recommendations are brought to the attention of any recipient of this research report.

Please refer to HTI's website www.equities.htisec.com for further information on HTI's organizational and administrative arrangements set up for the prevention and avoidance of conflicts of interest with respect to Research.

Non U.S. Analyst Disclosure: The HTI analyst(s) listed on the cover of this Research is (are) not registered or qualified as a research analyst with FINRA and are not subject to U.S. FINRA Rule 2241 restrictions on communications with companies that are the subject of the Research; public appearances; and trading securities by a research analyst.

分发和地区通知:

除非下文另有规定，否则任何希望讨论本报告或者就本项研究中讨论的任何证券进行任何交易的收件人均应联系其所在国家或地区的海通国际销售人员。

香港投资者的通知事项：海通国际证券股份有限公司 (“HTISCL”) 负责分发该研究报告，HTISCL 是在香港有权实施第 1 类受规管活动（从事证券交易）的持牌公司。该研究报告并不构成《证券及期货条例》（香港法例第 571 章）（以下简称“SFO”）所界定的要约邀请，证券要约或公众要约。本研究报告仅提供给 SFO 所界定的“专业投资者”。本研究报告未经过证券及期货事务监察委员会的审查。您不应仅根据本研究报告中所载的信息做出投资决定。本研究报告的收件人就研究报告中产生或与之相关的任何事宜请联系 HTISCL 销售人员。

美国投资者的通知事项: 本研究报告由 HTIRL, HSIPL 或 HTIJKK 编写。HTIRL, HSIPL, HTIJKK 以及任何非 HTISG 美国联营公司, 均未在美国注册, 因此不受美国关于研究报告编制和研究分析人员独立性规定的约束。本研究报告提供给依照 1934 年“美国证券交易法”第 15a-6 条规定的豁免注册的「美国主要机构投资者」(“Major U.S. Institutional Investor”)和「机构投资者」(“U.S. Institutional Investors”)。在向美国机构投资者分发研究报告时, Haitong International Securities (USA) Inc. (“HTI USA”) 将对报告的内容负责。任何收到本研究报告的美国投资者, 希望根据本研究报告提供的信息进行任何证券或相关金融工具买卖的交易, 只能通过 HTI USA。HTI USA 位于 130 West 42nd Street, FL 18, New York, NY 10036 USA, 电话+1 212-351-6052。HTI USA 是在美国于 U.S. Securities and Exchange Commission (“SEC”) 注册的经纪商, 也是 Financial Industry Regulatory Authority, Inc. (“FINRA”) 的成员。HTIUSA 不负责编写本研究报告, 也不负责其中包含的分析。在任何情况下, 收到本研究报告的任何美国投资者, 不得直接与分析师直接联系, 也不得通过 HSIPL, HTIRL 或 HTIJKK 直接进行买卖证券或相关金融工具的交易。本研究报告中出现的 HSIPL, HTIRL 或 HTIJKK 分析师没有注册或具备 FINRA 的研究分析师资格, 因此可能不受 FINRA 第 2241 条规定的与目标公司的交流, 公开露面和分析师账户持有的交易证券等限制。投资本研究报告中讨论的任何非美国证券或相关金融工具 (包括 ADR) 可能存在一定风险。非美国发行的证券可能没有注册, 或不受美国法规的约束。有关非美国证券或相关金融工具的信息可能有限。外国公司可能不受审计和汇报的标准以及与美国境内生效相符的监管要求。本研究报告中以美元以外的其他货币计价的任何证券或相关金融工具的投资或收益的价值受汇率波动的影响, 可能对该等证券或相关金融工具的价值或收入产生正面或负面影响。美国收件人的所有问询请联系:

Haitong International Securities (USA) Inc.

130 West 42nd Street, FL 18

New York, NY 10036

联系人电话: +1 212-351-6052

DISTRIBUTION AND REGIONAL NOTICES

Except as otherwise indicated below, any Recipient wishing to discuss this research report or effect any transaction in any security discussed in HTI’s research should contact the Haitong International salesperson in their own country or region.

Notice to Hong Kong investors: The research report is distributed by Haitong International Securities Company Limited (“HTISCL”), which is a licensed corporation to carry on Type 1 regulated activity (dealing in securities) in Hong Kong. This research report does not constitute a solicitation or an offer of securities or an invitation to the public within the meaning of the SFO. This research report is only to be circulated to “Professional Investors” as defined in the SFO. This research report has not been reviewed by the Securities and Futures Commission. You should not make investment decisions solely on the basis of the information contained in this research report. Recipients of this research report are to contact HTISCL salespersons in respect of any matters arising from, or in connection with, the research report.

Notice to U.S. investors: As described above, this research report was prepared by HTIRL, HSIPL or HTIJKK. Neither HTIRL, HSIPL, HTIJKK, nor any of the non U.S. HTISG affiliates is registered in the United States and, therefore, is not subject to U.S. rules regarding the preparation of research reports and the independence of research analysts. This research report is provided for distribution to “major U.S. institutional investors” and “U.S. institutional investors” in reliance on the exemption from registration provided by Rule 15a-6 of the U.S. Securities Exchange Act of 1934, as amended. When distributing research reports to “U.S. institutional investors,” HTI USA will accept the responsibilities for the content of the reports. Any U.S. recipient of this research report wishing to effect any transaction to buy or sell securities or related financial instruments based on the information provided in this research report should do so only through Haitong International Securities (USA) Inc. (“HTI USA”), located at 130 West 42nd Street, FL 18, New York, NY 10036 USA; telephone +1 212-351-6052. HTI USA is a broker-dealer registered in the U.S. with the U.S. Securities and Exchange Commission (the “SEC”) and a member of the Financial Industry Regulatory Authority, Inc. (“FINRA”). HTI USA is not responsible for the preparation of this research report nor for the analysis contained therein. Under no circumstances should any U.S. recipient of this research report contact the analyst directly or effect any transaction to buy or sell securities or related financial instruments directly through HSIPL, HTIRL or HTIJKK. The HSIPL, HTIRL or HTIJKK analyst(s) whose name appears in this research report is not registered or qualified as a research analyst with FINRA and, therefore, may not be subject to FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account. Investing in any non-U.S. securities or related financial instruments (including ADRs) discussed in this research report may present certain risks. The securities of non-U.S. issuers may not be registered with, or be subject to U.S. regulations. Information on such non-U.S. securities or related financial instruments may be limited. Foreign companies may not be subject to audit and reporting standards and regulatory requirements comparable to those in effect within the U.S. The value of any investment or income from any securities or related financial instruments discussed in this research report denominated in a currency other than U.S. dollars is subject to exchange rate fluctuations that may have a positive or adverse effect on the value of or income from such securities or related financial instruments. All inquiries by U.S. recipients should be directed to:

Haitong International Securities (USA) Inc.

130 West 42nd Street, FL 18

New York, NY 10036

Attn: Sales Desk at +1 212-351-6052

中华人民共和国的通知事项: 在中华人民共和国 (下称“中国”, 就本报告目的而言, 不包括香港特别行政区、澳门特别行政区和台湾) 只有根据适用的中国法律法规而收到该材料的人员方可使用该材料。并且根据相关法律法规, 该材料中的信息并不构成“在中国从事生产、经营活动”。本文件在中国并不构成相关证券的公共发售或认购。无论根据法律规定或其他任何规定, 在取得中国政府所有的批准或许可之前, 任何法人或自然人均不得直接或间接地购买本材料中的任何证券或任何实益权益。接收本文件的人员须遵守上述限制性规定。

加拿大投资者的通知事项: 在任何情况下该等材料均不得被解释为在任何加拿大的司法管辖区内出售证券的要约或认购证券的要约邀请。本材料中所述证券在加拿大的任何要约或出售行为均只能在豁免向有关加拿大证券监管机构提交招股说明书的前提下由 Haitong International Securities (USA) Inc. (“HTI USA”) 予以实施, 该公司是一家根据 National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations (“NI 31-103”) 的规定得到「国际交易商豁免」(“International Dealer Exemption”) 的交易商, 位于艾伯塔省、不列颠哥伦比亚省、安大略省和魁北克省。在加拿大, 该等材料在任何情况下均不得被解释为任何证券的招股说明书、发行备忘录、广告或公开发行。加拿大的任何证券委员会或类似的监管机构均未审查或以任何方式批准该等材料, 其中所载的信息或所述证券的优点, 任何与此相反的声明即属违法。在收到该等材料时, 每个加拿大的收件人均将被视为属于 National Instrument 45-106 Prospectus Exemptions 第 1.1 节或者 Securities Act (Ontario) 第 73.3(1) 节所规定的「认可投资者」(“Accredited Investor”), 或者在适用情况下 National Instrument 31-103 第 1.1 节所规定的「许可投资者」(“Permitted Investor”)。

新加坡投资者的通知事项: 本研究报告由 Haitong International Securities (Singapore) Pte Ltd (“HTISSPL”) 于新加坡提供。HTISSPL 是符合《财务顾问法》2001 (“FAA”) 定义的豁免财务

顾问,可(a)提供关于证券,集体投资计划的部分,交易所衍生品合约和场外衍生品合约的建议(b)发行或公布有关证券、交易所衍生品合约和场外衍生品合约的研究分析或研究报告。本研究报告仅提供给符合《证券及期货法》2001 第 4A 条项下规定的机构投资者。对于因本研究报告而产生的或与之相关的任何问题,本研究报告的收件人应通过以下信息与 HTISSPL 联系:

Haitong International Securities (Singapore) Pte. Ltd

10 Collyer Quay, #19-01 - #19-05 Ocean Financial Centre, Singapore 049315

电话: (65) 6713 0473

日本投资者的通知事项: 本研究报告由海通国际证券有限公司所发布,旨在分发给从事投资管理的金融服务提供商或注册金融机构(根据日本金融机构和交易法(“FIEL”))第 61 (1)条,第 17-11 (1)条的执行及相关条款)。

英国及欧盟投资者的通知事项: 本报告由从事投资顾问的 Haitong International Securities Company Limited 所发布,本报告只面向有投资相关经验的专业客户发布。任何投资或与本报告相关的投资行为只面对此类专业客户。没有投资经验或相关投资经验的客户不得依赖本报告。Haitong International Securities Company Limited 的分支机构的净长期或短期金融权益可能超过本研究报告中提及的实体已发行股本总额的 0.5%。特别提醒有些英文报告有可能此前已经通过中文或其它语言完成发布。

澳大利亚投资者的通知事项: Haitong International Securities (Singapore) Pte Ltd, Haitong International Securities Company Limited 和 Haitong International Securities (UK) Limited 分别根据澳大利亚证券和投资委员会(以下简称“ASIC”)公司(废除及过度性)文书第 2016/396 号规章在澳大利亚分发本项研究,该等规章免除了根据 2001 年《公司法》在澳大利亚为批发客户提供金融服务时海通国际需持有澳大利亚金融服务许可的要求。ASIC 的规章副本可在以下网站获取: www.legislation.gov.au。海通国际提供的金融服务受外国法律法规规定的管制,该等法律与在澳大利亚所适用的法律存在差异。

印度投资者的通知事项: 本报告由从事证券交易、投资银行及证券分析及受 Securities and Exchange Board of India (“SEBI”)监管的 Haitong Securities India Private Limited (“HTSIPL”)所发布,包括制作及发布涵盖 BSE Limited (“BSE”)和 National Stock Exchange of India Limited (“NSE”) (统称为「印度交易所」)研究报告。

(条款链接: <https://www.htisec.com/sites/all/themes/hitong/files/india/Terms%20and%20Conditions%20for%20Research%20Analyst.pdf>)

研究机构名称: Haitong Securities India Private Limited

SEBI 研究分析师注册号: INH000002590

地址: 1203A, Floor 12A, Tower 2A, One World Center

841 Senapati Bapat Marg, Elphinstone Road, Mumbai 400 013, India

CIN U74140MH2011FTC224070

电话: +91 22 43156800 传真: +91 22 24216327

合规和申诉办公室联系人: Prasanna Chandwaskar; 电话: +91 22 43156803; 电子邮箱: prasanna.chandwaskar@htisec.com

“请注意, SEBI 授予的注册和 NISM 的认证并不保证中介的表现或为投资者提供任何回报保证”。

本项研究仅供收件人使用,未经海通国际的书面同意不得予以复制和再次分发。

版权所有: 海通国际证券集团有限公司 2019 年。保留所有权利。

People's Republic of China (PRC): In the PRC, the research report is directed for the sole use of those who receive the research report in accordance with the applicable PRC laws and regulations. Further, the information on the research report does not constitute "production and business activities in the PRC" under relevant PRC laws. This research report does not constitute a public offer of the security, whether by sale or subscription, in the PRC. Further, no legal or natural persons of the PRC may directly or indirectly purchase any of the security or any beneficial interest therein without obtaining all prior PRC government approvals or licenses that are required, whether statutorily or otherwise. Persons who come into possession of this research are required to observe these restrictions.

Notice to Canadian Investors: Under no circumstances is this research report to be construed as an offer to sell securities or as a solicitation of an offer to buy securities in any jurisdiction of Canada. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by Haitong International Securities (USA) Inc., a dealer relying on the "international dealer exemption" under National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations ("NI 31-103") in Alberta, British Columbia, Ontario and Quebec. This research report is not, and under no circumstances should be construed as, a prospectus, an offering memorandum, an advertisement or a public offering of any securities in Canada. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon this research report, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. Upon receipt of this research report, each Canadian recipient will be deemed to have represented that the investor is an "accredited investor" as such term is defined in section 1.1 of National Instrument 45-106 Prospectus Exemptions or, in Ontario, in section 73.3(1) of the Securities Act (Ontario), as applicable, and a "permitted client" as such term is defined in section 1.1 of NI 31-103, respectively.

Notice to Singapore investors: This research report is provided in Singapore by or through Haitong International Securities (Singapore) Pte Ltd (“HTISSPL”). HTISSPL is an Exempt Financial Adviser under the Financial Advisers Act 2001 (“FAA”) to (a) advise on securities, units in a collective investment scheme, exchange-traded derivatives contracts and over-the-counter derivatives contracts and (b) issue or promulgate research analyses or research reports on securities, exchange-traded derivatives contracts and over-the-counter derivatives contracts. This research report is only provided to institutional investors, within the meaning of Section 4A of the Securities and Futures Act 2001. Recipients of this research report are to contact HTISSPL via the details below in respect of any matters arising from, or in connection with, the research report:

Haitong International Securities (Singapore) Pte. Ltd.

10 Collyer Quay, #19-01 - #19-05 Ocean Financial Centre, Singapore 049315

Telephone: (65) 6713 0473

Notice to Japanese investors: This research report is distributed by Haitong International Securities Company Limited and intended to be distributed to Financial Services Providers or Registered Financial Institutions engaged in investment management (as defined in the Japan Financial Instruments and Exchange Act ("FIEL") Art. 61(1), Order for Enforcement of FIEL Art. 17-11(1), and related articles).

Notice to UK and European Union investors: This research report is distributed by Haitong International Securities Company Limited. This research is directed at persons having professional experience in matters relating to investments. Any investment or investment activity to which this research relates is available only to such persons or will be engaged in only with such persons. Persons who do not have professional experience in matters relating to investments should not rely on this research. Haitong International Securities Company Limited's affiliates may have a net long or short financial interest in excess of 0.5% of the total issued share capital of the entities mentioned in this research report. Please be aware that any report in English may have been published previously in Chinese or another language.

Notice to Australian investors: The research report is distributed in Australia by Haitong International Securities (Singapore) Pte Ltd, Haitong International Securities Company Limited, and Haitong International Securities (UK) Limited in reliance on ASIC Corporations (Repeal and Transitional) Instrument 2016/396, which exempts those HTISG entities from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the financial services it provides to wholesale clients in Australia. A copy of the ASIC Class Orders may be obtained at the following website, www.legislation.gov.au. Financial services provided by Haitong International Securities (Singapore) Pte Ltd, Haitong International Securities Company Limited, and Haitong International Securities (UK) Limited are regulated under foreign laws and regulatory requirements, which are different from the laws applying in Australia.

Notice to Indian investors: The research report is distributed by Haitong Securities India Private Limited ("HSIPL"), an Indian company and a Securities and Exchange Board of India ("SEBI") registered Stock Broker, Merchant Banker and Research Analyst that, inter alia, produces and distributes research reports covering listed entities on the BSE Limited ("BSE") and the National Stock Exchange of India Limited ("NSE") (collectively referred to as "Indian Exchanges").

(Link to the Terms and Conditions document: <https://www.htisec.com/sites/all/themes/hitong/files/india/Terms%20and%20Conditions%20for%20Research%20Analyst.pdf>)

Name of the entity: Haitong Securities India Private Limited

SEBI Research Analyst Registration Number: INH000002590

Address : 1203A, Floor 12A, Tower 2A, One World Center

841 Senapati Bapat Marg, Elphinstone Road, Mumbai 400 013, India

CIN U74140MH2011FTC224070

Ph: +91 22 43156800

Details of the Compliance Officer and Grievance Officer : Prasanna Chandwaskar : Ph: +91 22 43156803; Email id: prasanna.chandwaskar@htisec.com

"Please note that Registration granted by SEBI and Certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors".

This research report is intended for the recipients only and may not be reproduced or redistributed without the written consent of an authorized signatory of HTISG.

Copyright: Haitong International Securities Group Limited 2019. All rights reserved.

<http://equities.htisec.com/x/legal.html>
